

The Tom Klootwijk Manifold

Exact bounded-query definition card – replacement v3.0

$$\text{TKM}_\beta(M) := (M \times Q, \Sigma, \mathcal{G}, \Phi, \text{Seg}_J, \Delta_K, \\ \mathbf{S}, \mathbf{C}, \mathbf{R}, \mathbf{T}, \mathbf{I}, \mathbf{L}; \beta), \quad \beta = (J, K, P, H, W).$$

$$x_i(t) = \text{Retr}_{\Phi(s_i, i, t)}(\Delta_{j(t), i}(t))$$

Exact query claim

$$T_\beta \leq c_\Phi + Jc_{\text{seg}} + Kc_\Delta + Hc_g + c_{\text{ret}} = c_\beta$$

$$T_\beta(\text{one indexed query}) = O(1)$$

The profile constants are fixed; overflow is rejected, split or assigned a new profile. For q results, work and output are $\Theta(q)$.

Correct lowercase little-o claim

$$B_N = B_0 + b_\nu \nu(N), \quad D_N = Nnb_x.$$

If $\nu(N) = o(N)$, then

$$B_N/D_N = o(1).$$

This is a normalized storage statement for fixed/sublinear novelty, not runtime.

Certified event

$$\text{Verified} = \text{IntervalCert} \wedge \text{Support} \wedge \text{Compatible} \\ \wedge (\epsilon \leq \mu) \wedge \text{SignSep}.$$

A field zero alone is not a transition or topology change. Endpoint sign separation certifies a zero only on a continuous, supported and compatible interval.

Factually correct only under: finite fixed profile; indexed entity/archive; unique segment; bounded cumulative corrections; certified field error; continuous interval certificate; explicit reset maps; no hidden dense output or unbounded history search.

Verdict: factually correct under the stated bounded profile. “Runtime is $o(1)$ ” is false; “one query is $O(1)$ ” and “encoded/dense ratio is $o(1)$ ” are the correct claims.